User Guide

Matrix Data Structure and Cocktail Sort Algorithm

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1. Overview

This project demonstrates two fundamental computer science concepts implemented in Python:

- **Matrix** – a two-dimensional abstract data type (ADT)
- **Cocktail Sort** – a bidirectional sorting algorithm

The integration example shows how Cocktail Sort can be applied to sort each row of a matrix.

2. Files Included

File	Description
`matrix.py`	Implements the Matrix class (create, set, display, transpose).
`cocktail_sort.py`	Contains the Cocktail Sort algorithm implementation.
`main.py`	Demonstrates both components, including timing results.
`user_guide.pdf`	User documentation.

3. System Requirements

Component	Version
Python	3.8 or later
OS	Windows / macOS / Linux
Libraries	Standard Python library only
IDE (optional)	VS Code, PyCharm, or IDLE

4. Installation Instructions

```
 git clone https://github.com/yourusername/matrix-cocktail-sort.git
cd matrix-cocktail-sort
python --version
python main.py
```

5. How to Use

Step 1 – Matrix Operations

```
=== Matrix Demonstration ===
Original Matrix:
[1, 2, 3]
[4, 5, 6]
[7, 8, 9]

Transposed Matrix:
[1, 4, 7]
[2, 5, 8]
[3, 6, 9]
```

Step 2 – Cocktail Sort Example

```
=== Cocktail Sort Demonstration ===
Original array: [5, 1, 4, 2, 8, 0, 3]
Sorted array:   [0, 1, 2, 3, 4, 5, 8]
Execution time: 0.007 ms
```

Step 3 – Sorting Each Row in a Matrix

```
=== Sorting Each Row in a Matrix with Cocktail Sort ===
Matrix before Sorting:
[75, 15, 98]
[9, 29, 22]
[25, 54, 67]

Matrix after Sorting Each Row:
[15, 75, 98]
[9, 22, 29]
[25, 54, 67]
```

6. Performance Example

Input	Output	Case	Time (ms)
[5,1,4,2,8,0,3]	[0,1,2,3,4,5,8]	Average	0.007
[0,1,2,3,4,5,8]	[0,1,2,3,4,5,8]	Best	0.001
[8,5,4,3,2,1,0]	[0,1,2,3,4,5,8]	Worst	0.012

7. Core Concepts

Concept	Description
Matrix (ADT)	Organizes data in two-dimensional form (rows and columns).
Cocktail Sort	Bidirectional sorting technique with forward and backward passes.
Integration	Sorting each matrix row independently using the algorithm.

8. Time Complexity Summary

Case	Complexity
Best	O(n)
Average	$O(n^2)$
Worst	$O(n^2)$
Space	O(1)

9. Customization

Modify `main.py` to test your own data:

```
 # Change Matrix size
matrix = Matrix(4, 4)

# Try a custom array
arr = [10, 4, 9, 1, 3]
```

10. Expected Output Summary

After running the program, you will see:

1. Matrix creation and transpose demonstration
2. Cocktail Sort demonstration with execution time
3. Matrix rows sorted using Cocktail Sort

11. References

- Python Official Documentation
- GeeksforGeeks – Cocktail Sort Explanation

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Designed for educational demonstration – Matrix & Cocktail Sort Project.